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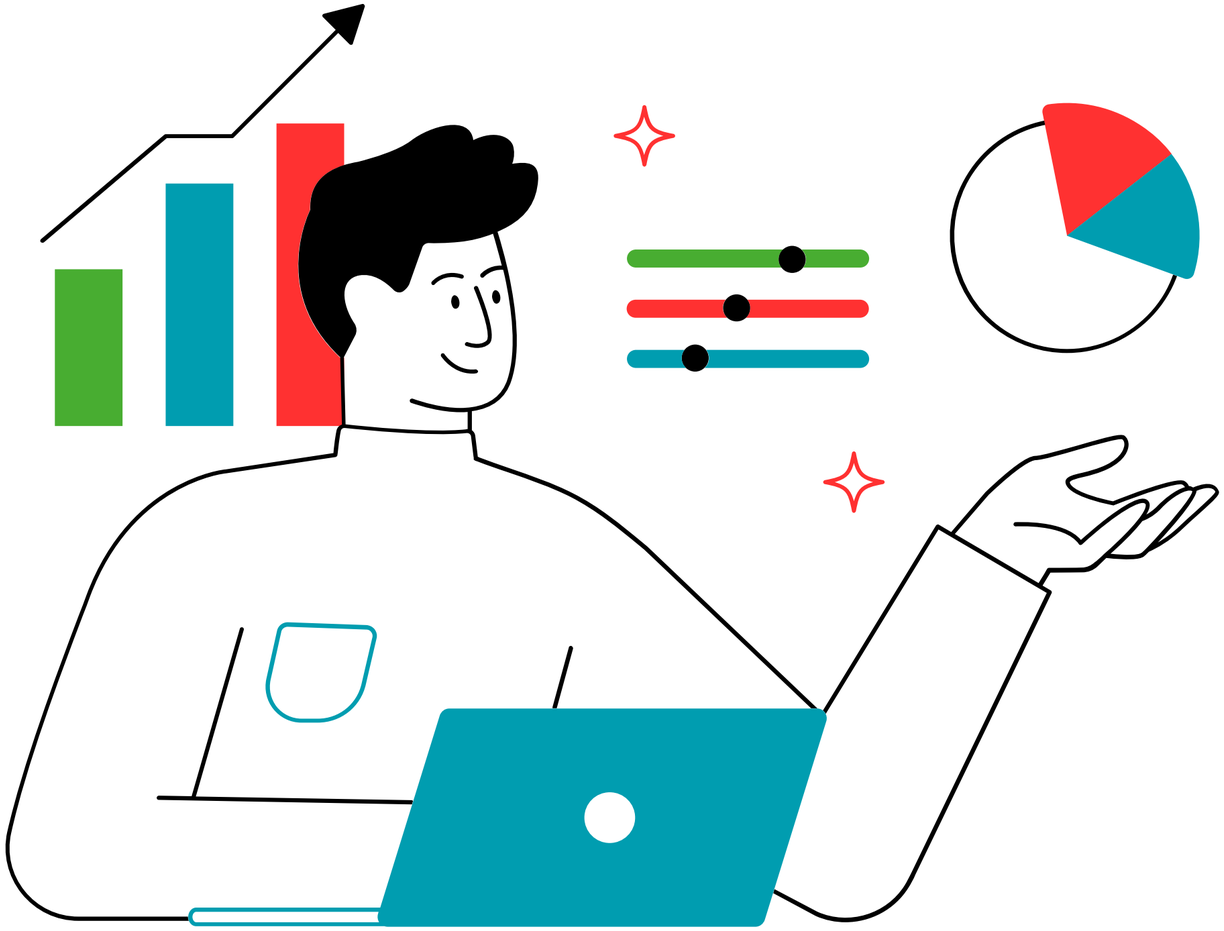
Data Analytics Study Guide

2026



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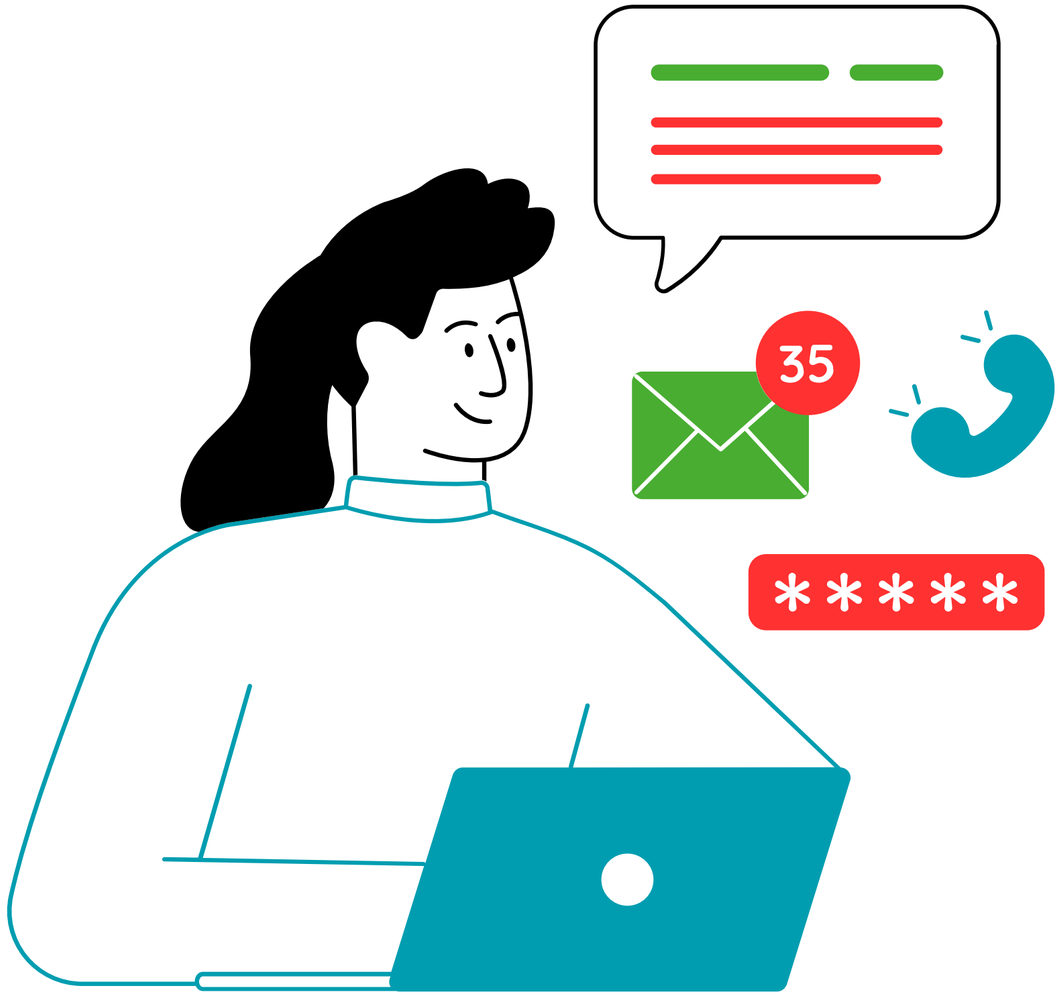

What is a Data Analyst?

Data Analyst is a professional responsible for **collecting, processing, and analyzing data** to help organizations make **informed business decisions.**

Data Analysts work with raw data from various sources, ensuring its accuracy and completeness before applying **statistical methods** and **analytical tools** to extract valuable **insights**. Using software like **Excel, SQL, or Python**, they identify trends, patterns, and correlations that **inform business strategies** and problem-solving.

Data Analysts also focus on **visualizing data** through **charts, graphs, and dashboards** to present complex information in an understandable format for decision-makers. Their role includes **reporting findings** and offering **actionable recommendations** based on **data insights,** collaborating with other departments to address business challenges and improve overall performance.

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Key Responsibilities of a Data Analyst

- **Collect & clean data:** Gather data from different sources and fix errors, missing values, duplicates, and formatting issues so it's analysis-ready.
- **Analyze data:** Use tools like Excel, SQL, and Python (plus basic statistics) to find trends, patterns, and performance drivers.
- **Visualize insights:** Build charts, dashboards, and simple visuals that make complex results easy to understand.
- **Report & present:** Turn findings into clear reports/slides with key insights and practical recommendations for decision-makers.
- **Solve business problems:** Translate business questions into analysis and propose actionable improvements based on evidence.
- **Collaborate with stakeholders:** Work with teams (marketing, finance, operations, etc.) to understand needs, define metrics, and deliver the right insights.
- **Maintain & monitor reporting:** Keep data pipelines, dashboards, and recurring reports accurate, updated, and consistent as the business changes.

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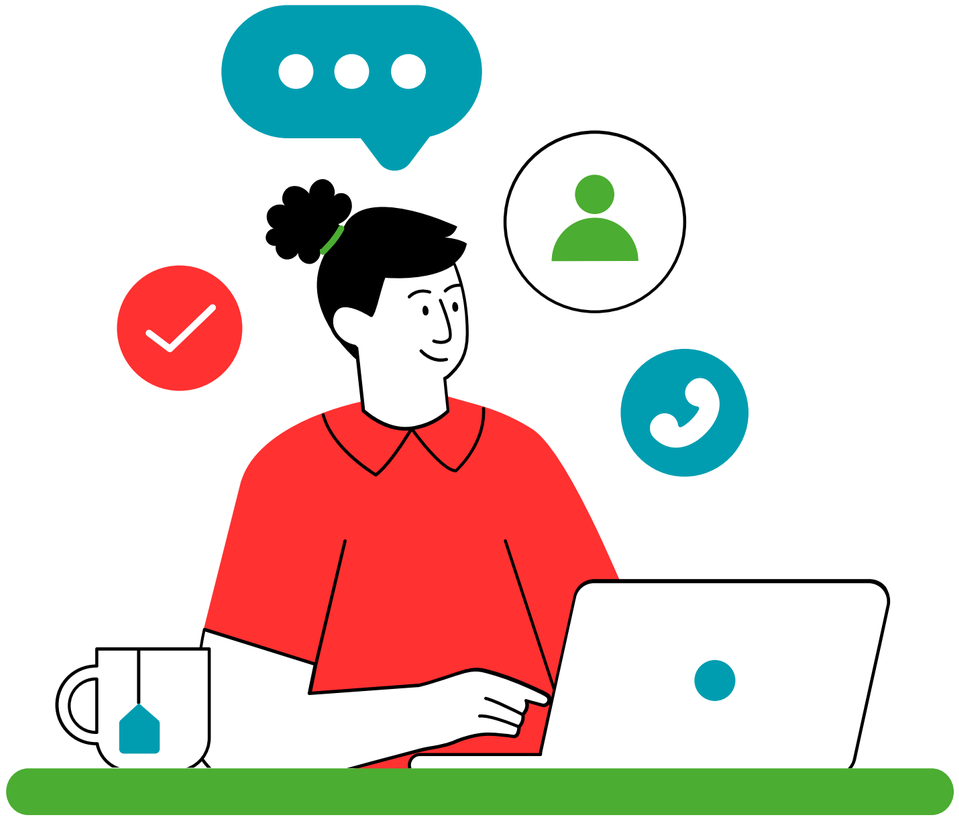
Essential Skills for Data Analysts



Technical Skills

- Data Manipulation &
- Cleaning
- Statistical Analysis
- Data Visualization
- Programming
- Database Management
- Data Integration and ETL
- Tools
- Version Control
- Business Intelligence Tools

Soft Skills

- Problem-Solving
- Attention to Detail
- Communication Skills
- Collaboration and Teamwork
- Time Management
- Critical Thinking
- Adaptability
- Curiosity and Learning Mindset
- Presentation Skills
- Ethical Thinking and Integrity

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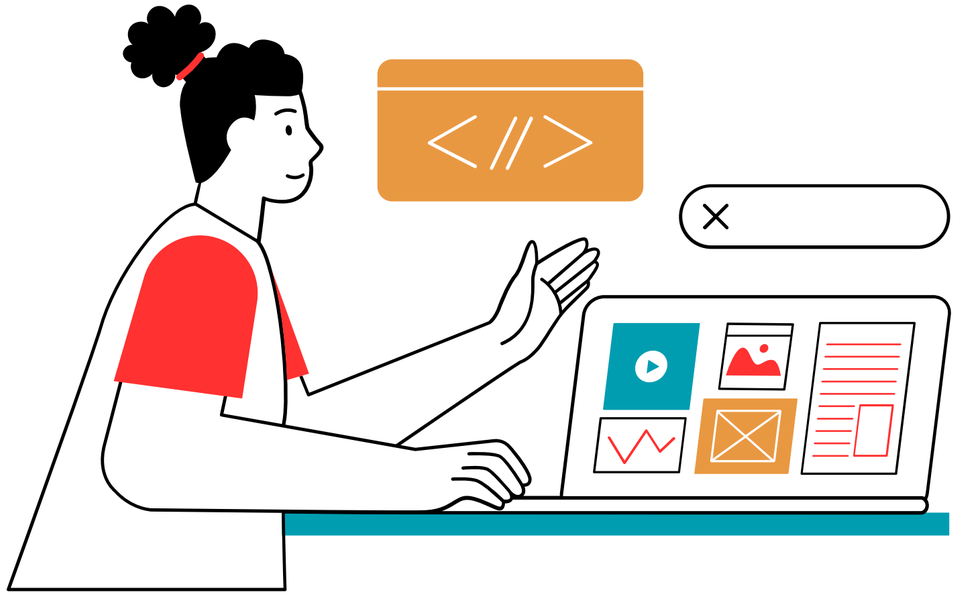

Tools & Technologies

Understanding Data Analytics Tools & Technologies

In our Data Analytics course, you'll learn that being a strong data analyst is not only about finding insights—it's also about knowing which tools to use and how to set them up correctly. Data doesn't arrive perfectly prepared, so you need to understand how an ETL pipeline works, meaning how data is extracted, cleaned, and loaded before it can be analyzed. You'll also learn how data is stored in a data warehouse, because this helps you know where to find the right tables, how to join data correctly, and how to avoid reporting the wrong numbers. That's why we train you on tools like Databricks, Snowflake, and Microsoft Excel, so you can confidently access real data, clean it properly, run analysis, and turn your results into insights that businesses can trust.

Data Cleaning:

- Microsoft Excel
- Google Sheets
- Structure Query Language (SQL)

Data Visualization:

- Microsoft Excel
- Google Sheets
- PowerBI

Data Presentation:

- Microsoft Power Point
- Google Slides
- Canva

Project Management

- Miro
- Canva
- Figma
- Microsoft Excel
- Google Sheets

Databases:

- Databricks
- Snowflake
- Microsoft SQL Server
- My SQL Workbench

Version Control:

- GitHub

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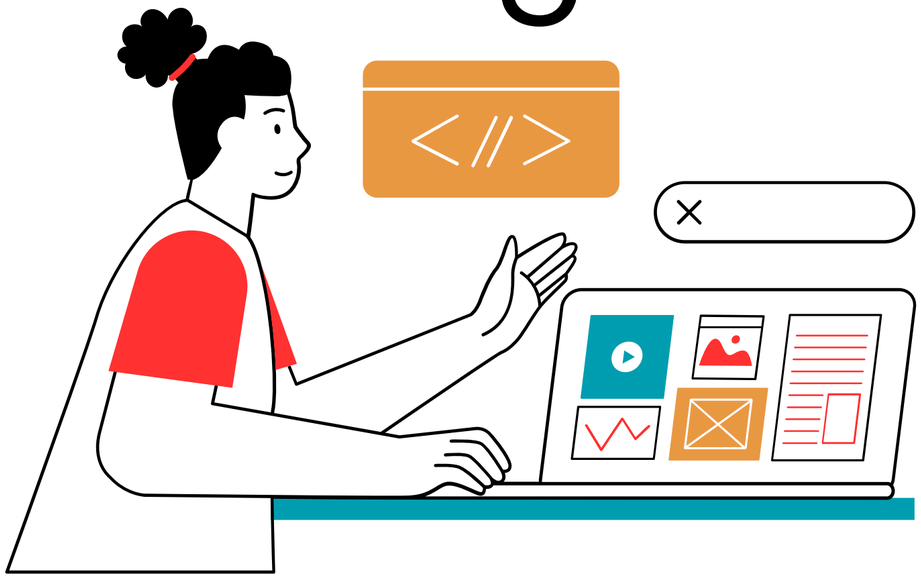
Access Course Tools & Technologies



Platform	Purpose	Platform Link
Databricks	Coding SQL	https://www.databricks.com/
Snowflake	Coding in SQL and building Databases	https://signup.snowflake.com/
BigQuery (Google Cloud)	Coding in SQL	https://console.cloud.google.com/bigquery
GitHub	Version control, sharing projects and building your data analytics portfolio.	https://github.com/
Canva	Building presentations	https://www.canva.com/templates
Miro	Project planning	https://miro.com/login/
Figma	Project planning	https://www.figma.com/design/UPUjGXCpd4S0Ok9ZgCmHEZ/Untitled?node-id=1-19&t=cmvFEdKGb4ZjCmgT-0
Looker Studio	Creating visual dashboards	https://lookerstudio.google.com/

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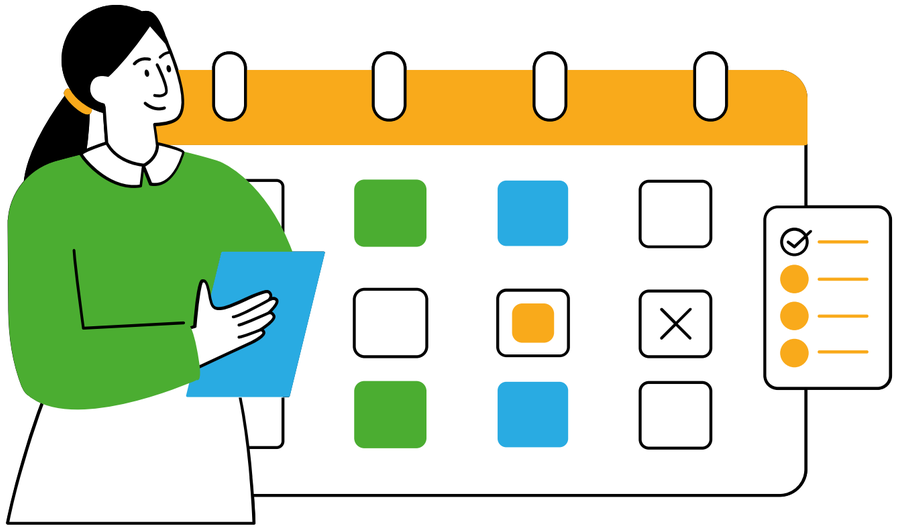
Data Analytics Program Laytout



Course Overview

- **Duration :** 3 Months
- **Level :** Beginners
- **Course Delivery :** Online
- **Tools Needed:** Laptop & WiFi
- **Additional Tools:** Notebook & Pen
- **Price:** R550 per month
- **Class Schedule:** Monday - Thursday
- **Class Time:** 19:00 - 20:00
- **Consultations:** Saturday & Sunday

Course Curriculum

- **Module 1:** Introduction to Databases
- **Module 2:** SQL Fundamentals
- **Module 3:** Advanced SQL
- Module 4: Data Visualization
- **Module 5:** Data Presentation
- Module 6: Project Management
- **Module 7:** Developing a Data Analytics Portfolio in Github

Skills You Will Gain

- Data analysis and interpretation
- Data cleaning, wrangling, and preparation using SQL
- Data aggregation and summarization
- techniques
- SQL for data manipulation and extraction
- Data Wrangling in Python
- Creating data visualizations in Excel
- Building interactive dashboards and
- reports in Power BI
- Developing a prof aessional portfolio to demonstrate your skills

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Data Analytics RoadMap

Introduction to Databases

Rational Databases (OLAP & OLTP), Data warehousing, Cloud Data systems, Data Management Systems

SQL Fundamentals

Learn SQL Basic syntax, including Aggregate functions, arithmetic operator, logical operators, etc

Advanced SQL

Learn Advanced SQL syntax, including window functions, stored procedure, etc

Data Visualization

PowerBI, Google Looker Studio, Microsoft Excel, google sheets, Snowflake, Databricks

Data Presentation

Google slides, Microsoft PowerPoint, Canva

Building a Portifolio

Build real world Data Analytics projects store it on Github.

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Introduction to SQL

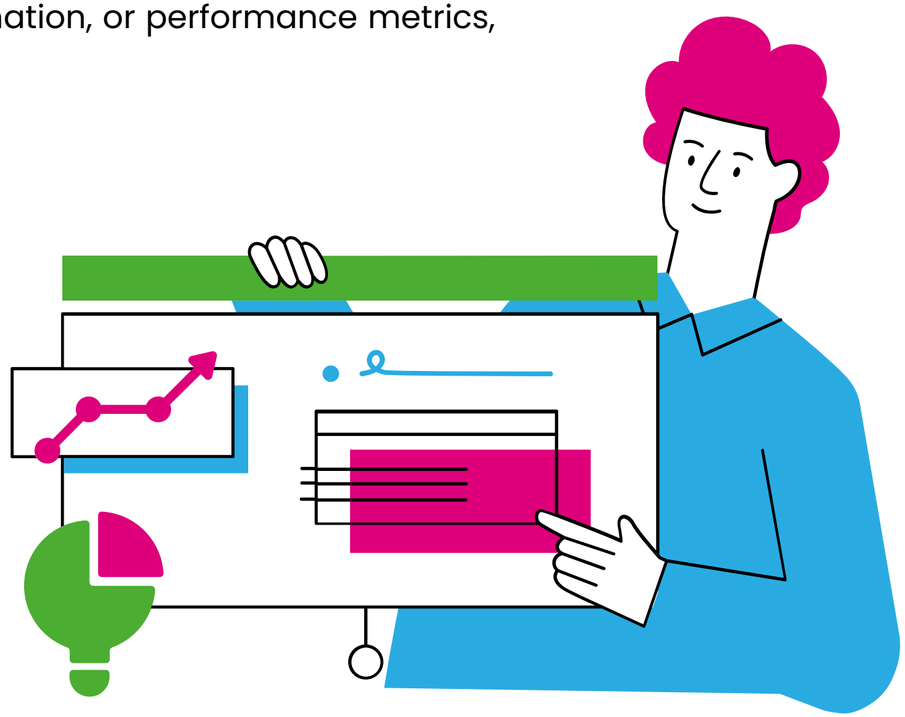
What is SQL?

SQL stands for Structured Query Language. It is the standard language used to store, retrieve, and manipulate data in relational databases. Whether you're analyzing sales data, customer information, or performance metrics, SQL is an essential tool in a data analyst's toolkit.

What can SQL do?

SQL With SQL, you can:

- Execute queries against a database
- Retrieve specific data from one or more tables
- Insert new records into a database
- Update existing records
- Delete unwanted records
- Create new databases and tables
- Create and manage stored procedures
- Create database views
- Set permissions on tables, procedures, and views



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Introduction to SQL

What is Relational Database?

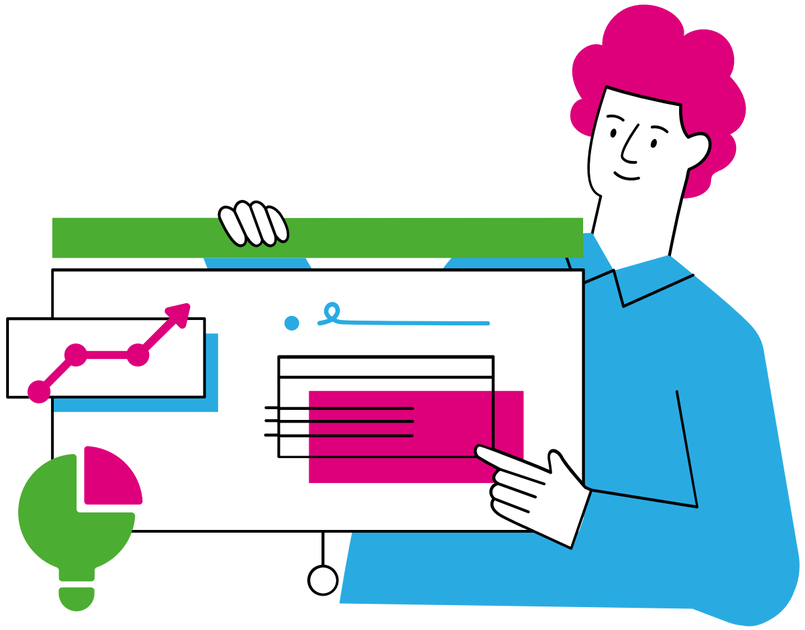
A relational database stores data in tables (rows and columns)

What is Relational Database Management System (RDBMS)?

An RDBMS is the software/platform that creates, stores, secures, and manages those relational tables and lets you run SQL reliably

Relational Database Management System (RDBMS)?

- **Organizes data reliably:** Stores data in structured tables with rules (schemas) so it stays clean and consistent.
- **Enables fast SQL querying:** Lets you retrieve, filter, join, and analyze data efficiently using SQL.
- **Protects and controls access:** Manages users, roles, and permissions so the right people see the right data.
- **Maintains data integrity:** Uses constraints and transactions to prevent duplicates, invalid values, and partial updates.
- **Supports scalability and performance:** Provides indexing/optimization and managed storage/compute to handle growing data and users.



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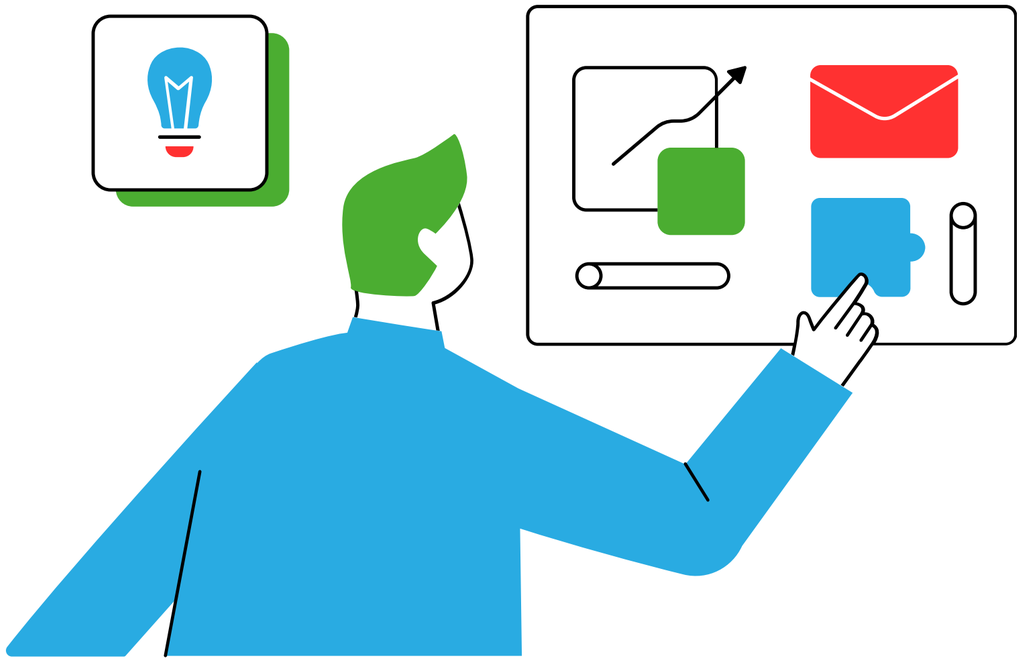
SQL Content Breakdown



SQL Fundamentals

This section will introduce basic SQL Syntax

1.1 SELECT Statements

- SELECT
- SELECT DISTINCT
- FROM
- AS (Aliases)

1.2 Filtering and Sorting

- WHERE Statement
- ORDER BY Clause
- SQL COMMENTS

1.3 SQL Operators

- Arithmetic Operators
- Bitwise Operators
- Comparison Operators
- Compound Operators
- Logical Operators

1.4 SQL Aggregate Functions s Filtering Grouped Data

- SUM()
- AVG()
- COUNT()
- MIN() and MAX()
- GROUP BY Clause
- HAVING Clause

1.5 Combining Tables (Joins and Unions)

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL OUTER JOIN
- SELF JOIN
- CROSS JOIN
- UNION Operator

1.6 Conditional Logic

- CASE Statements

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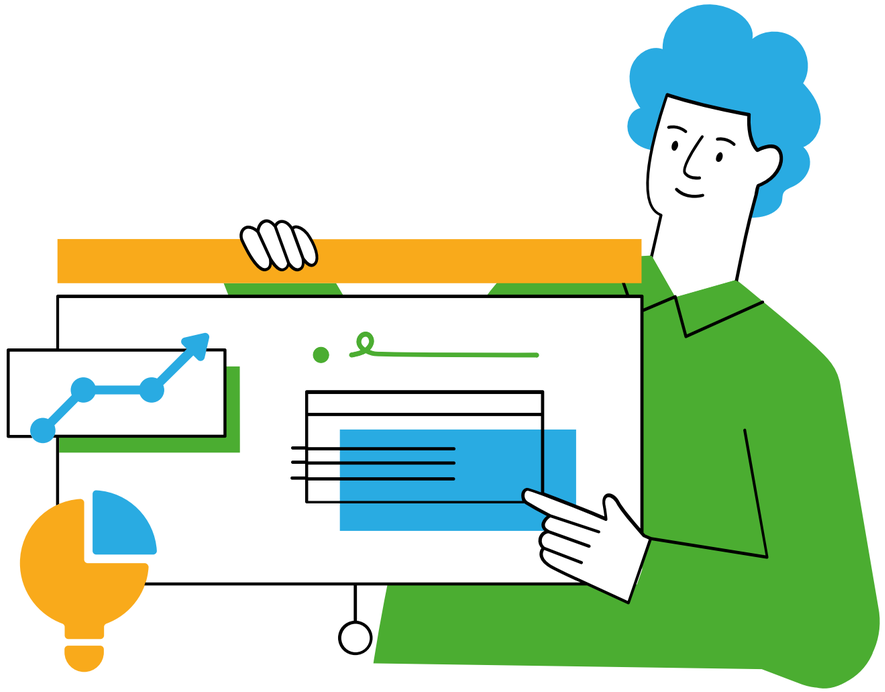
SQL Content Breakdown



Advanced SQL

This section will introduce Advanced SQL Syntax

2.1 Wildcards and Patterns
- SQL Wildcards (LIKE, %, _)

2.2 Filtering and Sorting
- NULL Functions
- DATE Functions

2.3 SQL Operators
- CREATE DATABASE
- DROP DATABASE
- BACKUP DATABASE

2.4 Table Operations
- CREATE TABLE
- DROP TABLE
- ALTER TABLE

2.7 Indexes
- CREATE INDEX
- DROP INDEX

2.5 Data Manipulation
- INSERT INTO
- SELECT INTO
- INSERT INTO SELECT
- UPDATE
- DELETE

2.6 Constraints
- UNIQUE Constraint
- CHECK Constraint
- DEFAULT Constraint

2.6 Constraints
- UNIQUE Constraint
- CHECK Constraint
- DEFAULT Constraint

2.8 Advanced Clauses
- EXISTS
- ALL Operator
- ANY Operator

2.9 Stored Procedures
- Creating and Using Stored Procedures

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Career Path & Opportunities

With **SQL + Excel + Power BI + data presentation**, you can qualify for roles across most industries because those skills are the "core toolkit" for turning raw data into business decisions. Here are strong career opportunities:

- **Data Analyst** – build reports/dashboards, track KPIs, explain trends and performance.
- **Business Intelligence (BI) Analyst** – design Power BI dashboards, data models, and executive reporting.
- **Reporting / MI (Management Information) Analyst** – scheduled reporting, scorecards, operational insights.
- **Product / Customer Insights Analyst** – analyze customer behavior, retention, funnels, segments, churn.
- **Marketing Analyst** – campaign performance, attribution, ROI, audience targeting insights.
- **Sales / Revenue Analyst** – pipeline, conversion, pricing analysis, sales performance dashboards.
- **Operations Analyst** – efficiency, turnaround times, cost tracking, process improvement using data.
- **Healthcare / Clinical Data Analyst** – patient flows, outcomes tracking, operational reporting (where applicable).
- **Government / Public Sector Data Analyst** – service delivery metrics, monitoring and evaluation reporting.
- **Big Data Analyst (entry path)** – if you add Databricks/Spark basics, you can analyze larger datasets.
- **Data Engineer / Analytics Engineer (junior path)** – if you add ETL + data modeling, you can build pipelines and curated datasets for BI.
- **Data Scientist (junior path)** – if you add statistics + Python + ML, you can move into predictive analytics.
- **Quantitative Analyst (limited path)** – usually needs strong statistics/programming, but your skills help as a foundation.
- **Finance Analyst / FP&A Analyst** – budgeting, forecasting, variance analysis, financial dashboards.
- **Supply Chain / Logistics Analyst** – inventory, demand trends, delivery performance, cost optimization.
- **HR / People Analytics Analyst** – hiring metrics, retention, performance, workforce planning.
- **Risk / Fraud Analyst** – detect unusual patterns, monitor risk indicators, reduce losses.

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Thank You

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